

Revision Notes for Class 8 Maths Chapter 2

Baudhāyana–Pythagoras Theorem

The Baudhāyana–Pythagoras Theorem is a foundational concept in geometry, with its origins rooted in ancient Indian mathematics. In this chapter, you learn how Indian mathematician Baudhāyana described methods to double the area of a square, halve a square, and relate the sides of a right-angled triangle. The notes summarise these mathematical ideas using clear step-by-step explanations, activities, and interesting observations.

Doubling and Halving a Square

The chapter begins with Baudhāyana’s question: “How can one construct a square having double the area of a given square?”. One might think that doubling the side length of a square would double its area. However, if the side of a square is doubled, its area becomes four times the original. Baudhāyana showed that the solution is to construct a square on the diagonal of the original square. The new square’s area is exactly double that of the original because the diagonal splits the area in such a way.

Similarly, to create a square half the area of a given one, instead of halving the side length, you construct a tilted square inside the original. This method ensures the new square’s area is half. Activities using paper and folding are suggested in the chapter for students to visualise and try these constructions.

Understanding the Hypotenuse in Isosceles Right Triangles

When you have a right triangle where the other two sides are equal, the length of the hypotenuse follows a special rule. If each small side is of length ‘a’, then the hypotenuse ‘c’ is $c = a\sqrt{2}$. This value of $\sqrt{2}$ lies between 1.414 and 1.415 and does not have a terminating decimal or a fractional form. The chapter provides bounds for $\sqrt{2}$ by squaring numbers like 1.4, 1.41, 1.414, and so on, showing that $\sqrt{2}$ is an irrational number.

The significance of the hypotenuse length and its calculation is fundamental for further work in right-triangle geometry, and the notes include worked examples with different side lengths to practice this understanding.

Baudhāyana-Pythagoras Theorem and Its Applications

The Baudhāyana-Pythagoras Theorem states that in a right-angled triangle with sides 'a' and 'b' and hypotenuse 'c', the square of the hypotenuse equals the sum of the squares of the other two sides: $a^2 + b^2 = c^2$. Baudhāyana mentioned that this rule works for any right triangle, not just those with equal smaller sides. This theorem allows us to calculate any missing side if the other two are known.

You also learn how to visually prove the theorem by rearranging squares and triangles on paper, and how combining or constructing new squares using sides of different lengths always leads back to this crucial relationship.

Integer Sided Right Triangles: Baudhāyana Triples

Certain sets of three positive integers, such as (3, 4, 5) and (5, 12, 13), satisfy $a^2 + b^2 = c^2$. These are called Baudhāyana triples, also known as Pythagorean triples. These triples are important for constructing right triangles with whole number sides – for example, a triangle with sides 3 cm, 4 cm, and 5 cm is a right-angled triangle.

There are infinitely many such triples; for example, by multiplying each element of a triple by the same number, you get new triples: (6, 8, 10), (9, 12, 15), etc. If a triple has no common factor except 1 (like (3, 4, 5)), it is called a primitive Baudhāyana triple.

- Primitive triples: Sides have no common divisor other than 1
- Non-primitive triples: Multiples of a primitive triple (e.g., (6, 8, 10))
- Triples can be generated by formulas involving sums of odd numbers

Further Applications and Special Problems

The theorem has creative applications, as shown through problems and puzzles. For example, finding the depth of a lake using a lotus stem as described by Bhāskarāchārya

uses the Pythagoras theorem directly. Many geometry problems, including finding diagonals of squares and rectangles, sides of rhombuses, and constructing special shapes, all rely on this core relationship.

Summary of Key Points

- Doubling or halving the area of a square cannot be done by simply doubling or halving the side length. Use the diagonal for doubling, and for halving, construct a smaller square inside the original with diagonals aligned.
- For right-angled triangles: $a^2 + b^2 = c^2$ holds true, where 'c' is the hypotenuse.
- In isosceles right triangles, the hypotenuse is $a\sqrt{2}$, and $\sqrt{2}$ is an irrational number between 1.414 and 1.415.
- Baudhāyana triples (e.g., (3, 4, 5)) provide integer solutions to the theorem and are building blocks for countless triangle shapes.
- Fermat's Last Theorem says that the equation $a^n + b^n = c^n$ has no integer solutions for $n > 2$.
- Baudhāyana's ideas are deeply rooted in the way we solve geometry and trigonometry questions even today.

Practice and Critical Thinking

- Try constructing a square with double, triple, or five times the area of a given square using only paper and folding.
- Given any two sides of a right triangle, use the theorem to find the third. Practice with side lengths like 3, 4, 5 or 5, 12, 13.
- List all Baudhāyana triples below 20. Identify which are primitive and which are not.
- Can a square with half the side length fill the original square exactly? Explain your answer based on the chapter.

Interesting Fact

Did you know? Baudhāyana described these rules and methods around 800 BCE, much before Pythagoras! His ancient Śulba-Sūtra is a rich source of geometric concepts still used today.

Chapter Summary Table

Concept	Key Point	Example
Doubling a Square	Use the diagonal to get twice the area	Original side = 1, New side = $\sqrt{2}$
Halving a Square	Construct a tilted square inside the original	Find new side using folding
Pythagoras Theorem	$a^2 + b^2 = c^2$ for right triangle	(3, 4, 5)
Baudhāyana Triples	Three integers solve $a^2 + b^2 = c^2$	(5, 12, 13)
Fermat's Last Theorem	No solution to $x^n + y^n = z^n$ if $n > 2$	–

Try This Puzzle

You have three boxes labelled RED, BLUE, and GREEN, but each is labelled incorrectly. If you can open only one box, how can you work out which box contains which colour?